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# From Uncorrelated Tracks to TLEs

*A Comprehensive Mathematical Pipeline with Provider-Specific Implementations*

LATEX → PDF · ~41 PAGES, AMS STYLE DOWNLOAD PDF

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## § Abstract

A complete, derivation-first treatment of the mathematical pipeline that turns uncorrelated observation tracks into the Two-Line Element sets that operational SDA systems publish. Every equation is derived or stated with full variable definitions; no step is skipped. The companion ambition is to document, in exhaustive detail, how the four major commercial and government SDA providers actually implement this pipeline in practice: the 18th Space Defense Squadron (formerly 18 SPCS), LeoLabs, Slingshot Aerospace, and ExoAnalytic Solutions.

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## § Scope and approach

Full pipeline; AMS article style; all available sources, including open literature (AMOS/AAS/AIAA papers, patents, SBIR abstracts, press), government documents where available (SDA, SSC TTPs, JFSCC, CSpOC SOPs), and direct vendor materials.

The construction is uncompromising on derivations. Where conventional treatments collapse "differentiate and solve" into a sentence, this monograph expands to the explicit Jacobian — partials of range, range-rate, and angles with respect to the six-element state — and tracks every approximation introduced along the way.

## § Initial Orbit Determination, complete

Laplace, Gauss, Gooding, Lambert (via Battin's universal-variable formulation, Gooding's iterative scheme, and Izzo's elegant rewrite), and Herrick–Gibbs are each derived from first principles, with worked numerical examples and discussion of which method to use under which observation geometry. The admissible region machinery — Tommei, Milani, Rossi — is presented in its modern form, including the connection to attributable orbit determination.

## § SGP4 and the TLE fit

The complete SGP4/SDP4 algorithm is reproduced with all constants — the WGS72 choice (not WGS84, despite the name of the latter being more familiar) is non-negotiable for parity with operational systems. The Vallado–Crawford–Hujsak–Kelso fit procedure converts a high-fidelity ephemeris back into a TLE by iteratively adjusting the mean elements so the SGP4 ephemeris matches the truth ephemeris in least-squares sense over the fit span.

The checksum mechanics (the last digit of each TLE line) are reproduced with the modulo-10 algorithm.

## § Provider deep-dives

For each of 18 SDS, LeoLabs, Slingshot, and ExoAnalytic: the architecture as it can be reconstructed from open materials, the sensor mix that feeds the pipeline, the filtering and IOD choices that distinguish them, the public-facing products (UCT catalogs, conjunction screening services, SSA-as-a-service APIs), and what is provably *not* in the open record. The intent is not to out the providers but to map the design space against the math in Parts I–II.

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# § Status & provenance

Final LaTeX is ~2800 lines; PDF is 41 pages. Produced in conversation 1349ca6e on 2026-04-09 after a research phase across 150+ publications. Required three `pdflatex` passes for full cross-reference resolution; `xcolor` with `[dvipsnames]` must precede `hyperref` to support mixed color specifications; stacked math accents (`\dot{\hat{\boldsymbol{L}}}`) need helper macros to dodge `\macc@adjust` errors.